FIRST SEMESTER B.VOC(HONOURS) SOUND ENGINEERING EXAMINATION

PSYCHOACOUSTICS – MCQ ANSWER KEY
SET A & SET B

SET – A

PART A (1 Mark Each)

1. Cerumen is produced in which part of the ear? → Ear canal

2. ITD cues are most useful for which frequency range? → Low frequencies

3. Leitmotif in film scoring means → Recurring musical theme tied to a character or idea

4. The sone scale is meant to be → Linear with loudness perception

5. Cultural context alters music emotion due to → Learned associations

6. A lossless format means → Original audio can be perfectly reconstructed

7. Structure responsible for balance → Semicircular canals

8. ILD cues are most effective for → High-frequency sounds

9. Spatial audio research focuses on → HRTF personalization and rendering accuracy

10. One octave above 440 Hz → 880 Hz

11. Binaural beat claims are → Mixed and need more research

12. To reduce reverberation → Introduce absorptive materials

13. Primary auditory cortex location → Temporal lobe (Auditory cortex)

14. Temporal masking includes → Backward masking

15. Masked high frequency in encoder → Discard or heavily quantize it

16. Critical bands describe → Strong spectral masking interaction

17. Music therapy targets mood via → Limbic system (mood & memory regulation)

18. Binaural rendering uses → HRTFs

19. Timbre depends on → Harmonic content & spectral envelope

20. Sequential grouping helps perceive → Melody

21. Skilled musicians have → Better time discrimination

22. Same SPL but different loudness due to → Frequency sensitivity of ear

PART B (2 Marks Each)

23. Cochlear nerve lesion effect → Sensorineural hearing loss

24. High cognitive load reduces → Speech intelligibility & working memory

25. Alarm design in noisy factory → Highly salient with irregular temporal pattern

26. Temporal resolution importance → Discriminates rapid consonant transitions

27. HRTF convolution ensures → Directional interaural differences

28. Perceptual lossy codec family → MPEG/MP3/AAC

29. Classroom amplification benefit → Improved signal-to-noise ratio

30. Cross-modal interaction means → Different senses influence each other

31. Noise control assertion → A is false, R is true

32. Conductive loss pattern → Air–bone gap present

33. Localization in reverberant room → Use DRR cues & head movement

34. Low latency in games → Synchronizes sound with player action

35. Ototoxic drugs cause → Permanent sensorineural loss

36. Shepard tone construction → Overlapping octave tones with fading envelopes

37. Directional mic assertion → A is false, R is true

38. Inner hair cell assertion → Both A and R are true; R explains A

39. Nucleus accumbens function → Reward and pleasure

SET – B

PART A (1 Mark Each)

1. Timbre depends on → Harmonic content & spectral envelope

2. Balance structure of inner ear → Semicircular canals

3. Encoder response to masked HF → Discard or heavily quantize it

4. Cerumen is produced in → Ear canal

5. Music therapy affects → Mood & memory

6. Binaural beat claims → Mixed, need more research

7. ITD cues most useful for → Low frequencies

8. Critical bands describe → Masking interaction

9. Leitmotif means → Recurring character theme

10. Temporal masking includes → Backward masking

11. Binaural rendering uses → HRTFs

12. Different loudness at same SPL due to → Ear frequency sensitivity

13. Spatial audio research studies → HRTF personalization

14. Auditory cortical processing area → Temporal lobe

15. ILD cues work best for → High frequencies

16. Sone scale is → Linear with loudness

17. One octave above 440 Hz → 880 Hz

18. Skilled musicians have → Better time discrimination

19. Cultural effect on emotion due to → Learned associations

20. Lossless format means → Perfect reconstruction

21. To reduce reverberation → Use absorptive materials

22. Sequential grouping perceives → Melody

PART B (2 Marks Each)

23. Directional microphone assertion → A is false, R is true

24. Low-latency in games needed for → Action–sound synchronization

25. High cognitive load reduces → Speech intelligibility & working memory

26. Localization in reverberant room → Use head movement & DRR cues

27. HRTF convolution ensures → Directional hearing cues

28. Shepard tone construction → Overlapping octave tones

29. Cross-modal interaction means → Senses influence each other

30. Lossy codec family → MPEG/MP3/AAC

31. Conductive hearing loss pattern → Air–bone gap present

32. Inner hair cell assertion → Both true & correctly explained

33. Classroom amplification helps by → Improving SNR

34. Psychoacoustic noise control assertion → A false, R true

35. Temporal resolution importance → Speech consonant discrimination

36. Cochlear nerve damage causes → Sensorineural hearing loss

37. Nucleus accumbens function → Reward & pleasure

38. Ototoxic drug effect → Permanent hair cell damage

39. Noisy factory alarm design → Highly salient irregular pattern